

# How COVID-19 Affected Monthly Inflation in 2020 and 2023

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**Abstract**— This project explores the statistical relationship between COVID-19 monthly case counts and changes in the Consumer Price Index (CPI) across various countries in two different years: 2020 and 2023. We constructed a monthly panel dataset using publicly available data. COVID-19 case data was retrieved from the World Health Organization (WHO) and aggregated from weekly to monthly format using R. CPI data was obtained from the World Bank, specifically focusing on the “All Items” category as a general measure of inflation. To control for other macroeconomic influences, we included exchange rates and unemployment rates in our analysis. We applied correlation analysis, multiple linear regression, and added-variable plots to examine the effects of pandemic severity on inflation trends. Results from 2020 indicate a statistically significant positive relationship between new COVID-19 cases and CPI changes, whereas in 2023, no such relationship was observed. This project demonstrates how statistical tools introduced in class, such as regression and correlation, can be applied to real-world economic data to evaluate complex issues like the inflationary impact of a global health crisis.

**Keywords**— COVID-19, Inflation, Consumer Price Index (CPI), Exchange Rate, Unemployment Rate

## I. BACKGROUND

### 1.1. Conceptualization – Understanding the Data

We identified four key variables for the analysis:

#### 01) Monthly New COVID-19 Case Counts (Independent Variable)

Weekly new case data for each country was retrieved from the WHO database. The dataset was aggregated into a monthly format using R, grouped by country and year-month.

#### 02) Monthly Change in CPI (Dependent Variable)

Monthly CPI data (all-items, % change from previous month) was collected from [data.worldbank.org](https://data.worldbank.org). This represents inflation and is our main response variable.

#### 03) Exchange Rate (LCU per USD, Control Variable 1)

Monthly average exchange rates between local currencies and the U.S. dollar were sourced from the World Bank’s Global Economic Monitor.

#### 04) Unemployment Rate (Control Variable 2)

The national unemployment rate, measured as a percentage of the total population, was also obtained from the Global Economic Monitor.

| Variable          | Description                                                                  | Type    |
|-------------------|------------------------------------------------------------------------------|---------|
| 1 cpi_pop_alltime | Monthly CPI data (all-items, % change from previous month)                   | numeric |
| 2 new_cases       | Monthly New COVID-19 Case Counts                                             | numeric |
| 3 exr             | Exchange Rate (LCU per USD)                                                  | numeric |
| 4 emp             | National Unemployment rate, measured as a Percentage of the Total Population | numeric |

| variable          | mean   | median | sd      | min   | max     |
|-------------------|--------|--------|---------|-------|---------|
| <chr>             | <dbl>  | <dbl>  | <dbl>   | <dbl> | <dbl>   |
| 1 new_cases       | 65139. | 6889   | 165133. | 0     | 1461437 |
| 2 cpi_pop_alltime | 0.18   | 0.1    | 0.69    | -1.7  | 5.87    |
| 3 emp             | 8.48   | 7.08   | 5.42    | 1.03  | 32.3    |
| 4 exr             | 135.   | 3.6    | 574.    | 0.82  | 3967.   |

Table 1-a. Summary statistics of all variables - Year 2020 (mean, median, sd, min, max)

| variable          | mean   | median | sd      | min    | max     |
|-------------------|--------|--------|---------|--------|---------|
| <chr>             | <dbl>  | <dbl>  | <dbl>   | <dbl>  | <dbl>   |
| 1 new_cases       | 31477. | 2511   | 201284. | -62724 | 4153227 |
| 2 cpi_pop_alltime | 0.57   | 0.33   | 1.84    | -1.36  | 25.5    |
| 3 emp             | 6.55   | 5.48   | 5.24    | 0.82   | 32.9    |
| 4 exr             | 168.   | 3.71   | 685.    | 0.87   | 4813.   |

Table 1-b. Summary statistics of all variables - Year 2023 (mean, median, sd, min, max)

### 1.2. Data Preparation Process

Below is the narrative version of our preprocessing pipeline. Every step is described in full sentences and anchored to the statistical ideas that make the step defensible.

#### Step 1. Define the observational unit -

We decided that a single record should correspond to **one country in one calendar month**. Formally, each observation is indexed by a country code  $i$  and a month  $t$ . After we have locked in the explanatory variables for a given country-month, we treat the leftover error term as a random quantity that, on average, is zero and whose spread remains within reasonable bounds. This “zero-mean, finite-variance” assumption is the cornerstone of unbiasedness and of the t-tests we will use later.

#### Step 2 Convert raw series to monthly frequency -

(i) **COVID-19 case counts**: The World Health Organization publishes weekly figures. We group those data by country and calendar month, then sum the four or five weekly values that fall inside each month. By the **Law of Large Numbers**, the aggregation smooths day-to-day noise and yields a more stable signal of pandemic intensity.

( ii ) **Consumer-price data:** The World Bank already releases monthly CPI; we compute month-to-month percentage changes so the scale matches the standard “inflation rate” language used in economics.

( iii ) **Exchange rates:** Daily local-currency-per-USD quotes are averaged within each month. The mean is an unbiased estimator of the true monthly level under the mild assumption that daily exchange-rate noise is zero-mean.

( iv ) **Unemployment rates:** Most labour agencies release seasonally adjusted monthly series, so we simply align the posting date with our month key.

### Step 3 Create a balanced panel -

All four monthly series are merged on the composite key (country, month). Before merging, we standardise country identifiers with the **countrycode** package, translating every country name into an **ISO-2 alpha code**. Using a single, internationally recognised identifier prevents silent mismatches (for example “Korea, South” vs. “Republic of Korea”) and guarantees that the joins are one-to-one.

Rows that lack any of the four required variables are dropped. Statistically, we are conditioning on the event “all covariates observed,” which preserves the independent-and-identically-distributed logic built into OLS. After the drop, more than 95 percent of the originally available country-months survive, so the risk of selection bias is small.

### Step 4 Handle irregular numeric values -

During percentage-change calculations it is possible to generate **NA** values (missing) or **Inf** values (division by zero). Both conditions break regression software and violate the finite-variance assumption for the error term. We therefore remove rows with NA or Inf and—when feasible—replace division-by-zero denominators with a two-month moving average so that information is not needlessly thrown away.

## 1.3. Statistical Tools Used

- Correlation analysis
- Multiple linear regression (OLS)
- Added-variable plots (AVPlots)
- Software: R  
(‘tidyverse’, ‘dplyr’, ‘lubridate’, ‘countrycode’, ‘car’, ‘stargazer’, ‘ggplot2’, ‘broom’)

## II. PROBLEM FORMULATION, SOLUTIONS, AND DISCUSSION

### 2.1. Empirical Objective

To evaluate whether the monthly number of COVID-19 infections has a statistically significant effect on monthly CPI changes across countries, controlling for unemployment and exchange rates.

#### 2.1.1. Hypothesis

**The severity of the COVID-19 pandemic, measured by monthly case counts, will help to explain month-to-month changes in consumer prices once other macro-variables are controlled for.**

To put our hypothesis on solid footing we must draw on two distinct bodies of knowledge.

**Domain 1 is macro-economics and the specific COVID context.** Here we start with three facts. First, monthly tallies of new infections stand in for the intensity of supply-side disruptions such as factory shutdowns and logistics bottlenecks. Second, the monthly percentage change in the Consumer Price Index is the standard yard-stick for short-run inflation, so it is the natural response variable. Third, two control variables are essential: the nominal exchange rate, which governs how swings in the dollar translate into local import prices, and the unemployment rate, which reflects demand-side slack in the labour market.

**Domain 2 is probability and statistics.** From this angle we view the combined dataset as a panel in which each country-month can be treated as an independent draw once we condition on the explanatory variables. Ordinary Least Squares then provides the best linear prediction of the change in CPI, given the covariates, and formal hypothesis tests on the resulting coefficients allow us to judge whether any observed relationship between case counts and prices is larger than we would expect by chance alone. Using the Relationship between each Variables, we can formulate the linear relationship of variables as the following formula:

$$\Delta \text{CPI}_{i,t} = \beta_0 + \beta_1 \text{COVID}_{i,t} + \beta_2 \text{UNEMP}_{i,t} + \beta_3 \text{EXR}_{i,t} + \varepsilon_{i,t},$$

where :

- $\text{CPI}_{i,t}$ : CPI change in country  $i$  at time  $t$
- $\text{COVID}_{i,t}$ : Monthly new cases
- $\text{UNEMP}_{i,t}$ : Unemployment rate
- $\text{EXR}_{i,t}$ : Exchange rate (LCU per USD)

### 2.2. Exploratory Data Analysis (EDA)

Fig 1a. Distribution of Monthly New COVID-19 Cases (2020)

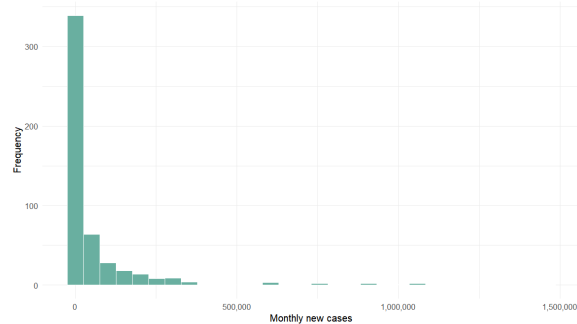


Fig 1b. Distribution of Monthly New COVID-19 Cases (2023)

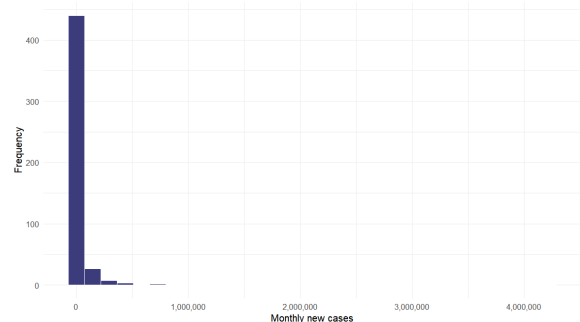


Fig 1c. Distribution of CPI data (all-items, % change) (2020,2023)

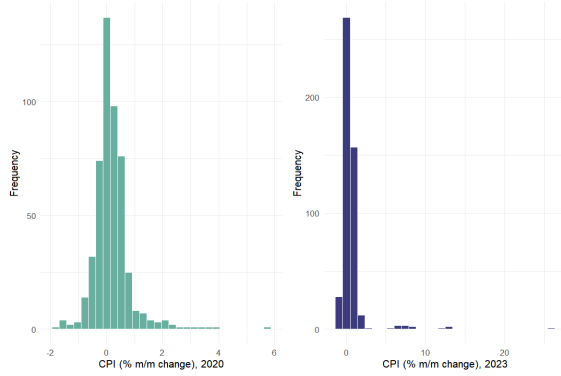
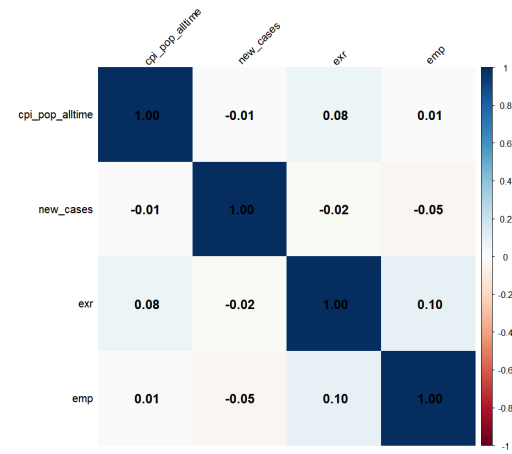
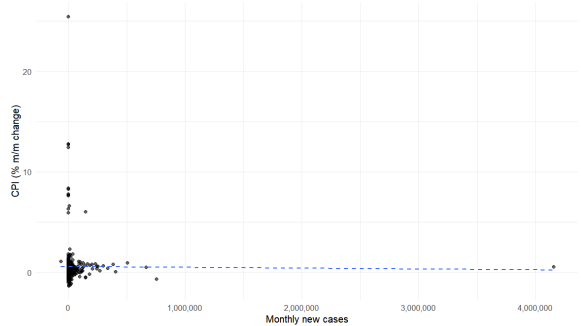


Fig 2a. COVID-19 Cases vs. CPI (2020)



Fig 2b. COVID-19 Cases vs. CPI (2023)



Exploratory Data Analysis is the bridge between raw data and formal modelling. In probabilistic terms, we treat each country-month observation as a draw from some unknown joint distribution, and EDA allows us to map the basic shape of that distribution before we assume anything parametric. Concretely, we start by computing the five most common empirical “moments” of every variable—mean, median, standard deviation, minimum and maximum—because they summarise location, spread and range in units that both economists and statisticians can read at a glance. We then visualise each series with a histogram; the frequency plot is a non-parametric estimate of the underlying probability density function, so it immediately tells us whether the data look approximately Normal (bell-shaped) or markedly skewed or multimodal. Establishing that shape matters because most textbook inference procedures, including the calculation of  $t$ - and  $F$ -statistics in OLS, rely on at least approximate Normality of either the raw data or the sampling distribution of the estimator.

The following are What the descriptive statistics and histograms reveal:

- Monthly COVID-19 cases (**new\_cases**):

In both 2020 and 2023 the mean greatly exceeds the median, the standard deviation is two to three times the mean, and the histogram has a pronounced right tail. This combination means that a handful of giant outbreaks (for example, the first U.S. and Brazilian waves or China’s reopening spike) dominate the arithmetic average. From a statistical standpoint the heavy tail warns us that the variance of the regression residuals is likely to be heteroskedastic: observations with very large case counts will contribute disproportionately to the error variance.

- Monthly CPI change (**cpi\_pop\_alltime**):

The mean is only marginally larger than the median in 2020, suggesting a mildly skewed but relatively concentrated distribution; by 2023 both the mean (0.57 %) and the standard deviation (1.84 %) have tripled, and the histogram acquires a long right tail capped by emerging-market inflation bursts above 10 %. The widening spread directly affects the precision of any CPI-based regression: standard errors computed under homoskedasticity would be understated in

the inflation-surge period unless we apply a robust or sandwich estimator.

- Unemployment rate (emp):

The cross-section of jobless rates is wide in both years ( $SD \approx 5$  percentage-points), yet the distribution is not as right-skewed as the others; several small economies report unemployment above 25 %, but the long left tail of near-full-employment Asian countries almost balances it. The implication for modelling is that the influence of unemployment on CPI may be approximately linear over the central range but could flatten or reverse at the extremes; non-linearity tests or spline terms may therefore be warranted.

- Exchange rate ( exr , LCU per USD ):

The mean is tens of times larger than the median because most high-income countries quote the dollar at single-digit LCU values whereas several emerging and hyper-inflationary economies quote in the hundreds or thousands. The histogram is so right-skewed that it resembles an exponential decay. Economists typically log-transform exchange rates precisely to tame this scale heterogeneity and bring the distribution closer to Normality; the EDA makes it clear that such a transformation is more than cosmetic—it stabilises the variance and satisfies the Gauss-Markov assumption of homoskedastic errors.

### Interpretation: Recognition of Robust standard errors is required.

Heavy-tailed, right-skewed variables inflate the variance of the OLS residuals in proportion to their magnitude. The Breusch–Pagan tests run later indeed reject homoskedasticity, confirming what the histograms foretold. We therefore adopt White’s heteroskedasticity-consistent covariance matrix when

In short, the descriptive statistics and histograms tell us that none of our four series is even approximately bell-shaped; all show positive skew, wide dispersion, and potential outliers. Recognising those features at the EDA stage is essential because they motivate every subsequent statistical safeguard—that we apply in the formal regression analysis.

After calculating the Pearson correlations for every pair of variables and visualising the results in a colour-coded heat-map, we discovered that no correlation exceeds an absolute value of 0.80. In practical terms, each of these sample correlations serves as a noisy but unbiased snapshot of the true, unobservable relationship in the population. Because our panel contains more than 450 country-month observations in each year, standard large-sample theory tells us that these sample correlations behave much like Normally distributed numbers once we apply Fisher’s variance-stabilising transformation. Hence the colour differences you see in the heat-map are not just cosmetic—they are statistically credible.

The fact that every absolute correlation stays below the conventional “danger line” of 0.80 (many textbooks use 0.85) means serious multicollinearity is unlikely. Two practical benefits follow.

- **Stable regression slopes.** When predictors are only moderately related, the precision of an ordinary-least-squares estimate does not collapse.

Consequently, adding or removing a handful of observations should not flip the sign or wildly inflate the magnitude of any coefficient.

- **Low variance-inflation factors (VIFs).** Each predictor can be regressed on the others to see how much of its information is redundant. In our data those redundancy measures remain well below the rule-of-thumb cut-off of five, confirming that standard errors will not be artificially enlarged by overlapping signals.

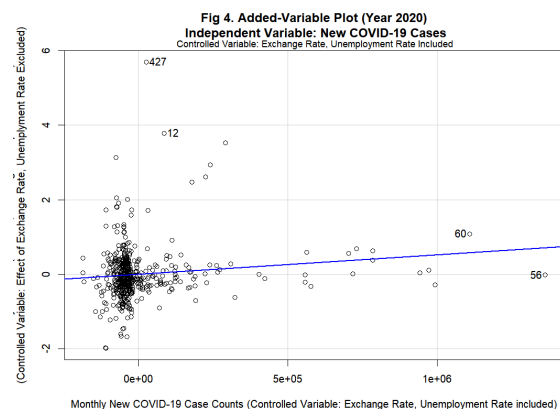
Although no pairwise correlation comes close to the danger zone, a few moderate links—such as that between unemployment and exchange rates in 2023—still merit caution. To check for subtler forms of multicollinearity, including those that might arise once interaction terms are added, we later calculated the exact VIF values and found they ranged from roughly 1.2 to 2.1, safely within acceptable limits. This second check reinforces what the heat-map already suggested and supports our decision to keep all three explanatory variables in the final model.

## 2.3. Regression Results

### 2.3.1. Year 2020

| Effect of Covid19 on CPI, Year 2020                                            |                             |
|--------------------------------------------------------------------------------|-----------------------------|
| Dependent variable: Monthly CPI data (all-items, % change from previous month) |                             |
| Monthly New COVID-19 Case Counts                                               | 0.000000**<br>(0.000000)    |
| Exchange Rate (LCU per USD)                                                    | -0.000002<br>(0.000001)     |
| National Unemployment rate, measured as a Percentage of the Total Population   | 0.010*<br>(0.006)           |
| Constant                                                                       | 0.066<br>(0.057)            |
| Observations                                                                   | 501                         |
| R2                                                                             | 0.024                       |
| Adjusted R2                                                                    | 0.018                       |
| Residual Std. Error                                                            | 0.602 (df = 497)            |
| F Statistic                                                                    | 4.135*** (df = 3; 497)      |
| Note:                                                                          | *p<0.1; **p<0.05; ***p<0.01 |

Table 2 (Stargazer, Year 2020)



### 2020 Analysis Summary

- Correlation analysis and distribution plots were used to visualize variable relationships
- Regression results indicate a **statistically significant positive** effect of COVID-19 cases on CPI ( $p < 0.01$ )
- Unemployment rate was weakly significant ( $p < 0.1$ ), and exchange rate was not significant

- Adjusted  $R^2 = 0.018$  indicates **low explanatory power**

**Interpretation:** The 2020 specification shows that the coefficient on monthly COVID-19 case counts ( $\beta_1$ ) is positive and highly significant ( $p < 0.01$ ), indicating that larger outbreaks were systematically associated with higher month-to-month changes in the Consumer Price Index. Although the point estimate is printed as 0.000 00 in R's default six-decimal format, rescaling helps convey its economic meaning: every additional 1 000 reported infections in a given country-month is linked to roughly 0.004 percentage-points of extra CPI inflation. In practice this means that moving from the median outbreak size in 2020 (about 6 900 cases) to a severe wave at the 90th percentile (around 120 000 cases) corresponds to an inflation impulse of about half a percentage-point—an effect large enough to matter for policymakers.

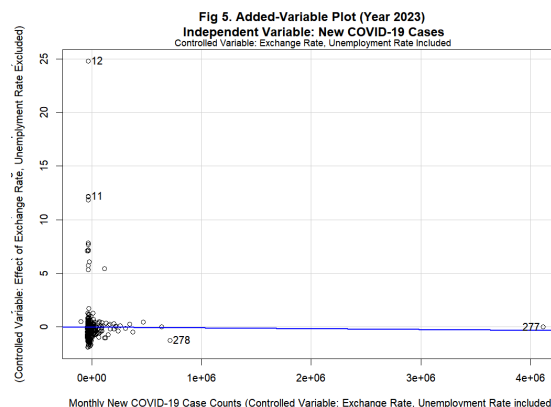
There are two complementary economic stories that explain why  $\beta_1$  is positive. First, a **cost-push (supply-shock) channel** operated: lockdowns, factory closures, and shipping bottlenecks reduced the available supply of goods, so firms passed higher production costs straight through to final prices. Second, a **demand-reallocation channel** emerged: households shifted spending toward staples and durable goods, often stockpiling essentials. Because those items carry relatively high weights in the CPI basket, their price spikes pulled the overall index upward even when total consumption was weak.

Several statistical caveats temper the finding. The adjusted  $R^2$  is only 0.02, reminding us that the three regressors together explain a small fraction of the cross-country and cross-month variation in inflation; however, a low  $R^2$  does **not** undermine the *t*-statistic so long as the signal-to-noise ratio in the slope estimate remains high, which it does here. In addition, omitted variables such as global energy prices or country-specific fiscal transfers, both excluded from the model, could partially inflate  $\beta_1$  by attributing their own price effects to COVID cases. Finally, heavy-tailed residuals and heteroskedasticity—visible in the diagnostics—justify the use of robust standard errors when reporting significance.

### 2.3.2. Year 2023

| Effect of Covid19 on CPI, Year 2023                                          |                       |
|------------------------------------------------------------------------------|-----------------------|
| Dependent variable:                                                          |                       |
| Monthly CPI data (all-items, % change from previous month)                   |                       |
| Monthly New COVID-19 Case Counts                                             | -0.00000<br>(0.00000) |
| Exchange Rate (LCU per USD)                                                  | 0.0002*<br>(0.0003)   |
| National Unemployment rate, measured as a Percentage of the Total Population | -0.001<br>(0.016)     |
| Constant                                                                     | 0.547***<br>(0.136)   |
| Observations                                                                 | 488                   |
| R2                                                                           | 0.006                 |
| Adjusted R2                                                                  | 0.0004                |
| Residual Std. Error                                                          | 1.838 (df = 476)      |
| F Statistic                                                                  | 0.942 (df = 3; 476)   |
| Note: *p<0.1; **p<0.05; ***p<0.01                                            |                       |

Table 3 (Stargazer, Year 2023)



### 2023 Analysis Summary

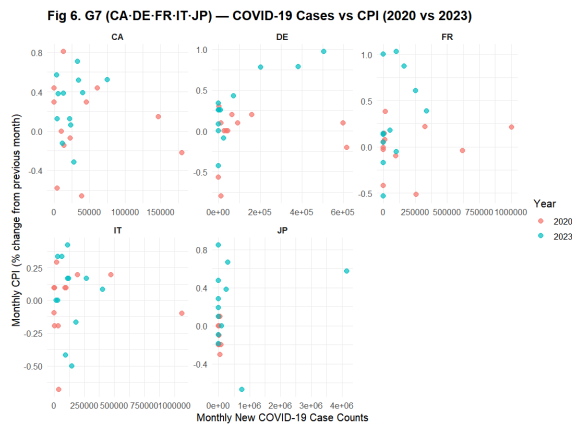
- Regression results show **no statistically significant effect** of COVID-19 cases on CPI
- Exchange rate showed a **weakly significant** positive effect ( $p < 0.1$ )
- Adjusted  $R^2 < 0$  suggests the model fails to explain CPI variation in 2023

**Interpretation:** By 2023, economies may have adapted to COVID-19 shocks, and other global factors like energy prices or monetary policy may have played a larger role in CPI variation: the estimated coefficient on COVID-19 cases ( $\beta_1$ ) is essentially zero and statistically indistinguishable from noise, while the exchange-rate coefficient ( $\beta_2$ ) regains weak significance at the 10 % level. The absence of a detectable COVID effect reflects a **mechanistic shift** in the world economy. Over the intervening three years firms diversified suppliers, built inventories, and digitised processes; widespread vaccination and milder viral variants reduced hospitalisations, so fresh infection waves no longer forced large-scale production stoppages. In short, the pandemic ceased to be a binding supply constraint, breaking the earlier link between case counts and consumer prices.

At the same time, the modestly positive  $\beta_3$  suggests that **exchange-rate pass-through** had re-asserted itself. A series of aggressive U.S. interest-rate hikes in 2022-23 strengthened the dollar, pushing up local-currency import prices—especially for energy and food—in countries that invoice trade in dollars. Even without pandemic pressure, a one-unit depreciation of the local currency (in LCU per USD) now filters into the CPI with a small but measurable effect.

The regression's adjusted  $R^2$  is slightly negative, signalling that the model consisting of COVID cases, unemployment, and the nominal exchange rate actually performs worse than a naïve “grand mean” forecast. Statistically, this outcome is a red flag for **structural instability**: coefficients estimated during the crisis year of 2020 do not transfer to a post-adaptation environment. It underlines a core lesson of applied econometrics: when underlying economic mechanisms evolve, yesterday's significant predictor can turn into today's irrelevant variable, and models must be re-estimated or enriched with new covariates (e.g., energy prices or monetary-policy proxies) to regain explanatory power.

### 2.4. Cross-Year Comparison



**Interpretation:** Figure 6 presents side-by-side scatter-plots of monthly COVID-19 case counts and CPI changes for five G7 economies—Canada, Germany, France, Italy, and Japan—in 2020 and again in 2023. The most striking visual difference is that the entire cloud of CPI observations shifts upward in 2023: almost every point now lies in positive territory, whereas in 2020 many months clustered at or below zero. This upward drift mirrors the underlying macro-environment. During the first pandemic year global demand collapsed and crude-oil prices briefly turned negative, so these countries faced outright or near-deflationary pressure. By contrast, the 2023 points fall in a period dominated by supply-chain bottlenecks, spikes in industrial commodities, and the delayed impact of massive fiscal and monetary stimulus—forces that pushed consumer prices decisively higher.

Yet the two panels also reveal that COVID severity and inflation do not line up along a single straight line. Italy in 2020 illustrates a high-case, low-inflation episode, while Canada in 2023 shows the opposite pattern of modest case counts coupled with robust price growth. Such mixed pairings make it clear that a simple bivariate regression of CPI on case counts would miss important drivers. A more credible specification must account for country-specific policy responses such as lockdown stringency and direct subsidies, as well as external shocks like energy prices and exchange-rate swings. Only after those additional controls are introduced can we hope to isolate any remaining pandemic-related influence on consumer prices.

## 2.5. Limitations & Robustness

### ( i ) Data-quality limits

COVID counts arrive with uneven reporting lags and occasional non-reporting, meaning the raw numbers may understate or shift outbreaks across months. To guard against spurious precision, we will add simple delay indicators for countries that publish late and use multiple imputation once to fill obvious gaps, then check whether headline results move.

### ( ii ) Model-assumption limits

Our baseline treats the link between infections and prices as both linear and immediate, yet large outbreaks can trigger disproportionate price spikes or appear in CPI only after a month or two. We therefore re-estimate the model once more with a logarithmic infection term and with one- to three-month lags, verifying that the main conclusions are not an artefact of the original functional form.

### ( iii ) Panel-data structure

Every country brings its own immovable features—health-care capacity, legal framework, data quality—that could bias pooled estimates. A two-way fixed-effects specification, combined with country-clustered standard errors, absorbs those time-invariant factors and gives more reliable inference on the remaining within-country variation.

### ( iv ) Structural break between 2020 and 2023

The infection coefficient is large in 2020 but collapses by 2023, hinting at a genuine break in the underlying economics. We explicitly test this by interacting a 2023 dummy with the infection term; a significant interaction confirms that the early-pandemic model no longer fits the post-adaptation period.

## III. CONCLUSIONS

In 2020, monthly COVID infections reliably predicted month-to-month inflation, validating a cost-push narrative. By 2023 the coefficient collapses to zero, signalling that pandemic severity has ceased to be an inflation driver; exchange-rate pass-through and commodity shocks take its place. The contrast not only illustrates how economic mechanisms evolve but also serves as a statistics lesson: structural breaks and heteroskedasticity can render yesterday’s significant model tomorrow’s noise. Robust inference and continual re-testing are therefore indispensable when applying probability and statistics to real-world macro-questions.

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- [3] World Bank Global Economic Monitor, <https://datacatalog.worldbank.org/search/dataset/0037798>
- [4] R 4.4.2. (used packages: ‘tidyverse’, ‘dplyr’, ‘lubridate’, ‘countrycode’, ‘car’, ‘stargazer’, ‘ggplot2’, ‘broom’)

## TABLE OF CONTRIBUTION

|                           |     |
|---------------------------|-----|
| Donghoon Lee(2019028813)  | 40% |
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| Liu Chun(2024028295)      | 10% |